

Dominant Arm Identification with Mixing and Recycling Observed Samples

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Crossing CDFs can change the selected arm

Classical objective

$$a_{\text{mean}} = \arg \max_{a \in [K]} \mathbb{E}[R_a].$$

Optimizes long-run expected reward.

Dominance objective

$$G_k = \left\{ r : \mathbb{P}(R_k > r) \geq \max_{j \neq k} \mathbb{P}(R_j > r) \right\},$$

$$D_a = \sum_{k=1}^K \int_{G_k} F_k(r) dP_a(r), \quad a^* = \arg \max_a D_a.$$

The score compares arm a with the strongest distribution at each reward level.

Biomedical simulation: the selected arm changes

	Largest mean	Largest dominance score
Biomedical simulation	Arm 1: $\mathbb{E}[R_1] = 0.413$	Arm 2: $D_2 = 0.420$

The criterion captures competitive performance across the reward distribution, including information that the mean hides.

Learning every CDF from partial feedback

At round t , the learner observes only $R_{a_t,t}$, while every score D_a depends on all K reward CDFs.



Doubly robust CDF update, with $\pi_{k,t} = \mathbb{P}(\tilde{a}_t = k)$

$$\tilde{F}_{k,t}(x) = \tilde{F}_{k,t}(x) + \frac{\mathbf{1}\{\tilde{a}_t = k\}}{\pi_{k,t}} \left(\mathbf{1}\{R_{\tilde{a}_t,t} \leq x\} - \tilde{F}_{k,t}(x) \right).$$

Sample reuse shares information across arms, while the propensity correction controls dependence and imputation error.

Source: Sim and Kim (2026), Section 3.1; Theorem 5 and Corollary 6.

Upper and lower bounds match in gap dependence

Let $a^* = \arg \max_a D_a$ and $\Delta_a = D_{a^*} - D_a$. DRDSE estimates the scores and eliminates arms after their confidence intervals separate.

Uniform CDF estimation

$$\max_{k \in [K]} \left\| \hat{F}_{k,t} - F_k \right\|_{\infty} = \tilde{O}\left(t^{-1/2}\right).$$

Information-theoretic lower bound

For suitable hard instances, every successful algorithm requires at least

$$\sum_{a \neq a^*} \frac{D_a^2}{12\Delta_a^2} \log \frac{1}{4\delta}$$

observations.

DRDSE upper bound

With probability at least $1 - 21\delta$, DRDSE uses

$$\tilde{O}\left(K + \sum_{a \neq a^*} \Delta_a^{-2}\right)$$

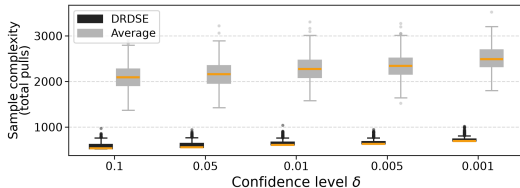
observations.

The upper and lower bounds have the same Δ_a^{-2} dependence up to logarithmic factors.

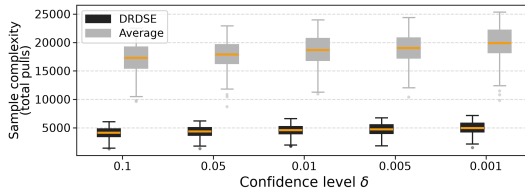
Source: Sim and Kim (2026), Theorems 2, 7, and 9.

DRDSE reduces sample complexity in both settings

Gaussian setting



Biomedical treatment-response setting



Both methods identify the true dominant arm in all 500 runs. Across confidence levels, Average uses 2.77–4.07 times as many median pulls for Gaussian rewards and 3.91–4.17 times as many for simulated treatment responses.

Source: Sim and Kim (2026), Figures 2 and 4.

Appendix: detailed slide map

- ▶ Partial-feedback estimation challenge
- ▶ Dominance-score error decomposition and DRDSE algorithm
- ▶ Implications and research direction
- ▶ Selected proofs and supporting lemmas

Challenge: every score depends on every CDF

At round t , the learner chooses a_t and observes only $R_{a_t,t}$.

$$(\dots, \dots, R_{a_t,t}, \dots, \dots)$$

What the score requires

- ▶ All K reward CDFs
- ▶ The empirical regions $\hat{G}_k$
- ▶ Every dominance score $\hat{D}_a$

What standard estimation gives

$$\left\| \hat{F}_{k,\text{emp}} - F_k \right\|_{\infty} = \tilde{O} \left(N_{k,t}^{-1/2} \right).$$

Rarely selected arms learn slowly.

The bottleneck is simultaneous uniform estimation of all K CDFs under partial feedback.

Source: Sim and Kim (2026), Section 3.1, pp. 3–4.

Error decomposition for the dominance score

For arm a and sample s , define $\epsilon_{a,s}$, $\check{\epsilon}_{a,s}$ (partition discrepancy), and $\bar{\epsilon}_{a,s}$ (DR estimator error).

Decomposition of the estimation error (Eq. 2)

$$|\hat{D}_{a,t} - D_a| \leq \left| \frac{1}{N_{a,t}} \sum_s \epsilon_{a,s} \right| + \left| \frac{1}{N_{a,t}} \sum_s \check{\epsilon}_{a,s} \right| + \left| \frac{1}{N_{a,t}} \sum_s \bar{\epsilon}_{a,s} \right|.$$

Lemma 8 (Partition discrepancy bound)

If $\max_k \|F_k - \hat{F}_{k,t}\|_\infty \leq \gamma_t$, then $\left| \frac{1}{N_{a,t}} \sum_s \check{\epsilon}_{a,s} \right| \leq 2\gamma_t$.

- ① Concentration term (Azuma): $\leq \sqrt{2 \log(Kt^2/\delta)/N_{a,t}}$.
- ② Partition bias (Lemma 8): $\leq 2\gamma_t$.
- ③ DR estimator error (Corollary 6): $\leq \gamma_t$.

Summing the three bounds gives exactly the confidence radius $\beta_{a,t} = 3\gamma_t + \sqrt{2 \log(Kt^2/\delta)/N_{a,t}}$ used by DRDSE.

DRDSE algorithm

- 1 Initialize the active set

$$\mathcal{A}_1 = [K].$$

- 2 Collect exploration blocks when the current exploration level is insufficient.

- 3 Update

$$\hat{F}_{k,t}, \quad \hat{G}_{k,t}, \quad \hat{D}_{a,t}.$$

- 4 Remove arms whose confidence intervals no longer overlap with a stronger candidate.

Elimination rule

Eliminate arm b if some active arm a satisfies

$$\hat{D}_{a,t} - \hat{D}_{b,t} > 2\beta_t.$$

Stopping rule

Stop when one arm remains:

$$|\mathcal{A}_t| = 1.$$

The algorithm converts simultaneous CDF estimation into a standard confidence-based identification procedure.

Source: Sim and Kim (2026), Algorithm 1, p. 7.

Implications and research direction

Implication

Crossing reward distributions can change the selected treatment even when one arm has the largest mean. The method makes that distribution-level decision learnable under bandit feedback, with applications wherever realized rankings determine outcomes.

Research direction

The next step is to sharpen finite-sample constants and reduce the current $O(Kt)$ computation. I also want to extend dominance identification to contextual or safety-constrained settings.

This work turns a distribution-level criterion into an identifiable bandit objective and shows when the distinction changes decisions.

Source: Sim and Kim (2026), Section 5, Appendix C.3, and pp. 31–32.

Appendix: selected proofs

- A Proof — Theorem 2 (sample complexity lower bound)
- B Proof — Lemma 3 (mixed Gram matrix spectrum bound)
- C Proof — Theorem 9 (DRDSE sample complexity upper bound)
- D Proof — Lemma 10 (two-point construction, used in A)
- E Proof — Theorem 4 (mixed estimator error bound)
- F Proof — Theorem 5 (doubly robust estimator error bound)
- G Proof — Lemma 11 (importance weighting bound, used in B)
- H Statement — Lemma 12 (matrix Hoeffding inequality, used in B, F)
 - I Statement — Lemmas 13–14 (pointwise-to-uniform bounds, used in E, F)
- J Statement — Lemmas 15, 16, 19 (threshold, Bretagnolle–Huber, martingale)

Appendix A: proof of Theorem 2 (lower bound)

- Reorder $D_1 > \dots > D_K$ and define $\Delta_a := D_1 - D_a$ for $a > 2$. Let $\hat{D}_{a,n_a}$ denote an estimator using n_a samples to estimate D_a . The event

$$\bigcap_{a=2}^K \left\{ |\hat{D}_{a,n_a} - D_a| \leq \frac{\Delta_a}{2} \right\} \cap \left\{ |\hat{D}_{1,n_1} - D_1| \leq \frac{\Delta_2}{2} \right\}$$

is a necessary condition to identify the best dominance score arm.

- For the event $\mathcal{B}$ of failing the identification,

$$\mathbb{P}(\mathcal{B}) \geq \max_{a=2,\dots,K} \mathbb{P}\left(|\hat{D}_{a,n_a} - D_a| > \frac{\Delta_a}{2}\right).$$

- If $n_a \leq \frac{D_a^2}{12\Delta_a^2} \log \frac{1}{4\delta}$ for some a , by Lemma 10,

$$\max_{a=2,\dots,K} \mathbb{P}\left(|\hat{D}_{a,n_a} - D_a| > \frac{\Delta_a}{2}\right) \geq \delta.$$

- Thus if the total number of samples is less than $\sum_{a=2}^K \frac{D_a^2}{12\Delta_a^2} \log \frac{1}{4\delta}$, the probability of failing the identification is greater than δ .

Source: Sim and Kim (2026), Appendix D.1.

Appendix B: proof of Lemma 3 (Gram matrix)

Recall $\tilde{V}_t := \sum_{s \in [t] \cap \mathcal{E}} w_s e_{a_s} e_{a_s}^\top + \sum_{s \in [t] \setminus \mathcal{E}} w_s \tilde{X}_{a_s, s} \tilde{X}_{a_s, s}^\top$.

- By the matrix Bernstein bound, with probability at least $1 - 2\delta/t^2$,

$$\left\| \sum_{s \in [t] \setminus \mathcal{E}} w_s \tilde{X}_{a_s, s} \tilde{X}_{a_s, s}^\top - \sum_{s \in [t] \setminus \mathcal{E}} w_s (I_K/K + e_{a_s} e_{a_s}^\top) \right\|_2 \leq 8\sqrt{2|[t] \setminus \mathcal{E}| \log \frac{Kt^2}{\delta}}.$$

- By construction of Ψ (applying Lemma 12 with $K = 1$) and $t \geq 15 \log \frac{Kt^2}{\delta}$, each with probability at least $1 - 2\delta/t^2$,

$$|[t] \cap \Psi| \geq \frac{t}{3}, \quad |[t] \setminus \Psi| \leq \frac{7}{2} \log \frac{Kt^2}{\delta}.$$

- Combining the two bounds via the importance weight bound (Lemma 11), with probability at least $1 - 4\delta/t^2$,

$$\left\| \tilde{V}_t - (|[t] \cap \Psi| + |[t] \setminus \mathcal{E} \cap \Psi|) I_K \right\|_2 \leq \sqrt{2t \log \frac{Kt^2}{\delta}} (9 + 8\sqrt{K}) \leq \frac{t}{6}.$$

- Taking a union bound over all steps above, with probability at least $1 - 10\delta/t^2$,

$$\frac{t}{6} I_K \preceq \tilde{V}_t \preceq \frac{13t}{6} I_K.$$

Appendix C1: proof of Theorem 9 — setup

- ▶ To satisfy the exploration condition $|[t] \cap \mathcal{E}| \geq 8K \log \frac{Kt^2}{\delta}$, by the threshold lemma (Lemma 15) we need at least

$$32K \left(1 + \log \frac{16K\sqrt{K}}{\sqrt{\delta}e} \right)$$

samples — this produces the $64K \log \frac{16K\sqrt{K}}{\sqrt{\delta}e}$ term in Theorem 9.

- ▶ Order $D_1 > \dots > D_K$. Arm $a > 2$ is eliminated once $\hat{D}_{1,T} > \hat{D}_{a,T} + \beta_{a,T} + \beta_{1,T}$. By the convergence of $\hat{D}_{a,t}$ (Theorem 7), with probability at least $1 - 10\delta$, this holds once

$$\beta_{a,T} + \beta_{1,T} \leq \frac{\Delta_a}{2}.$$

- ▶ Because the algorithm selects all arms in the undetermined set $\mathcal{A}_t$, the condition

$$3\gamma_T + \sqrt{\frac{2 \log(T^2/\delta)}{N_{a,T}}} \leq \frac{\Delta_a}{4}$$

implies the elimination.

Appendix C2: proof of Theorem 9 — conclusion

- Since $\gamma_T \leq 1926 \sqrt{\frac{3 \log(Kt^2/\delta)}{2N_{a,T}}}$, the condition from C1 is implied by

$$N_{a,T} \geq \frac{32 \cdot (3 \cdot 1926 + 1)^2}{\Delta_a^2} \log \frac{KT^2}{\delta}.$$

- Summing over $a > 2$ gives $T \geq \sum_{a=2}^K \frac{32 \cdot (3 \cdot 1926 + 1)^2}{\Delta_a^2} \log \frac{T^2}{\delta}$, which (applying Lemma 15 again to resolve the implicit T^2) is implied by, with probability at least $1 - 21\delta$ overall,

$$T \geq \left(1 + \sum_{a=2}^K \frac{4 \cdot 32 \cdot (3 \cdot 1926 + 1)^2}{\Delta_a^2}\right) \log \frac{\sqrt{K} \sum_{a=2}^K \frac{64 \cdot (3 \cdot 1926 + 1)^2}{\Delta_a^2}}{e\sqrt{\delta}}.$$

Source: Sim and Kim (2026), Appendix D.5.

Appendix D: proof of Lemma 10

For $v \in \{-1, 1\}$ and $\sigma_a := D_a/2$, construct noise η_v with

$$p_+(v) := \frac{1}{2} + \frac{\epsilon}{\sigma_a} v, \quad p_-(v) := \frac{1}{2} - \frac{\epsilon}{\sigma_a} v,$$

so that $\sum_{k=1}^K F_k(R_a) \mathbf{1}(R_a \in \mathcal{G}_k) = D_a + \eta_v$ under $\mathbb{P}_{a,v}$.

- ▶ Reducing estimation to hypothesis testing and applying the Bretagnolle–Huber inequality,

$$\sup_{v \in \{-1, 1\}} \mathbb{P}_v^{(n)}(|D_a - \hat{D}_{a,n}| > \epsilon) \geq \frac{1}{4} \exp(-n \text{KL}(\mathbb{P}_1^{(n)}, \mathbb{P}_{-1}^{(n)})).$$

- ▶ The KL divergence of the two-point family satisfies

$$\text{KL}(\mathbb{P}_1^{(n)}, \mathbb{P}_{-1}^{(n)}) = \frac{2\epsilon}{\sigma_a} \log \frac{1 + 2\epsilon/\sigma_a}{1 - 2\epsilon/\sigma_a} \leq \frac{12\epsilon^2}{\sigma_a^2}.$$

- ▶ Setting $n \leq \frac{\sigma_a^2}{12\epsilon^2} \log \frac{1}{4\delta}$ therefore ensures

$$\sup_{v \in \{-1, 1\}} \mathbb{P}_v^{(n)}(|D_a - \hat{D}_{a,n}| > \epsilon) \geq \delta.$$

Appendix E: proof of Theorem 4 (mixed estimator)

By the triangle inequality, $e_k^\top (\tilde{\mathbf{F}}_t(x) - \mathbf{F}(x))$ splits into a direct term and a recycled-mixing term.

- Using $\tilde{V}_t \succeq (t/6)I_K$ (Lemma 3), the dimension-free martingale bound (Lemma 19), and the uniform bound (Lemma 13), with probability at least $1 - 2\delta/(Kt^2)$ the direct term satisfies

$$\sup_{x \in \mathbb{R}} \left| \sum_{s \in [t] \cap \mathcal{E}} w_s \eta_s(x) e_k^\top \tilde{V}_t^{-1} e_{a_s} + \sum_{s \in [t] \setminus \mathcal{E}} w_s \eta_s(x) e_k^\top \tilde{V}_t^{-1} \tilde{X}_s \right| \leq 102 \sqrt{\frac{3}{t} \log \frac{Kt^2}{\delta}}.$$

- Because each explored round is reused at most $Kt/|[t] \cap \mathcal{E}|$ times, applying Lemma 19 and Lemma 13 again, with probability at least $1 - 2\delta/(Kt^2)$ the recycled-mixing term satisfies

$$\sup_{x \in \mathbb{R}} \left| \frac{1}{K} \sum_{s \in [t] \setminus \mathcal{E}} \sum_{k=1}^K m_{k,s} w_s \eta_{\tau_s+k-1}(x) e_k^\top \tilde{V}_t^{-1} \tilde{X}_s \right| \leq 294 \sqrt{\frac{1}{|[t] \cap \mathcal{E}|} \log \frac{2Kt^2}{\delta}}.$$

- Combining the two bounds, for all $k \in [K]$ with probability at least $1 - \frac{14\delta}{t^2}$,

$$\|\tilde{F}_{k,t} - F_k\|_\infty \leq 102 \sqrt{\frac{3}{t} \log \frac{Kt^2}{\delta}} + 294 \sqrt{\frac{1}{|[t] \cap \mathcal{E}|} \log \frac{2Kt^2}{\delta}}.$$

Appendix F: proof of Theorem 5 (DR estimator error)

Write $U_{k,s} := 1 - \frac{\mathbf{1}(\tilde{a}_s(\tau_s) = k)}{\mathbb{P}(\tilde{a}_s(\tau_s) = k \mid \mathcal{F}_{\tau_s}^-)}$. As in Lemma 3, with probability at least $1 - \delta/t^2$, $|[t] \cap \Psi| \geq t/3$, so

$$|\hat{F}_{k,t}(x) - F_k(x)| \leq \frac{3}{t} \left| \sum_{s=1}^t w_s \eta_{k,s}(x) \right| + \frac{3}{t} \left| \sum_{s \in \Psi \cap [t]} U_{k,s} \right| \cdot |\tilde{F}_{k,t}(x) - F_k(x)|.$$

- By the Hoeffding bound and the pointwise-to-uniform lemma (Lemma 14), with probability at least $1 - 2\delta/(Kt^2)$ the direct term satisfies

$$\sup_{x \in \mathbb{R}} \frac{3}{t} \left| \sum_{s=1}^t w_s \eta_{k,s}(x) \right| \leq 9 \sqrt{\frac{1}{t} \log \frac{Kt^3}{\delta}}.$$

- By the matrix concentration inequality (Lemma 12) applied to $U_{k,s}$, with probability at least $1 - 4\delta/(Kt^2)$,

$$\frac{1}{t} \left| \sum_{s \in \Psi \cap [t]} U_{k,s} \right| \leq 4 \sqrt{\frac{2K}{t} \log \frac{Kt^2}{\delta}}.$$

- Combining the two bounds, for all $k \in [K]$ with probability at least $1 - \frac{7\delta}{t^2}$,

$$\|\hat{F}_{k,t} - F_k\|_\infty \leq 9 \sqrt{\frac{1}{t} \log \frac{Kt^3}{\delta}} + 12 \sqrt{\frac{2K}{t} \log \frac{Kt^2}{\delta}} \|\tilde{F}_{k,t} - F_k\|_\infty.$$

Appendix G: proof of Lemma 11 (importance)

Lemma 11. Given $\delta \in (0, 1)$, for $t \geq 8K \log \frac{Kt^2}{\delta}$, suppose $\{X_{k,s}\}_{k \in [K], s \in [t]}$ are non-negative matrices with $\|\sum_{k=1}^K X_{k,s}\|_2 \leq b$ for every s . Then with probability at least $1 - 4\delta/t^2$,

$$\left\| \sum_{s \in [t]} w_s X_{a_s, s} - \sum_{s \in [t] \cap \Psi} \sum_{k=1}^K X_{k,s} \right\|_2 \leq b|[t] \setminus \Psi| + 4b\sqrt{2Kt \log \frac{Kt^2}{\delta}}.$$

- ▶ On Ψ^c , since $X_{a_s, s}$ is nonnegative definite, $\|\sum_{s \in [t] \cap \Psi^c} X_{a_s, s}\|_2 \leq |[t] \cap \Psi^c|b$.
- ▶ On Ψ , rewriting via the inverse selection probability and defining the martingale difference

$$D_s := \sum_{k=1}^K \left(\frac{\mathbf{1}(\tilde{a}_s(\tau_s) = k)}{\mathbb{P}(\tilde{a}_s(\tau_s) = k \mid \mathcal{F}_{\tau_s}^-)} - 1 \right) X_{k,s}, \quad \|D_s\|_2 \leq 2Kb.$$

- ▶ By the matrix Hoeffding inequality (Lemma 12), with probability at least $1 - 4\delta/t^2$,

$$\left\| \sum_{s \in [t]} D_s \right\|_2 \leq 2b\sqrt{2Kt \log \frac{Kt^2}{\delta}}, \quad \left\| \sum_{s \in [t] \setminus \Psi} D_s \right\|_2 \leq 2b\sqrt{2Kt \log \frac{Kt^2}{\delta}}.$$

Appendix H: Lemma 12 (matrix Hoeffding inequality)

Lemma 12. Let $\{M_\tau\}_{\tau=1}^t$ be a matrix-valued stochastic process adapted to a filtration $\{\mathcal{F}_\tau\}$, and let $\Delta_\tau := M_\tau - \mathbb{E}[M_\tau \mid \mathcal{F}_{\tau-1}]$ be the martingale difference matrices, with $\|\Delta_\tau\|_2 \leq b$ almost surely. Then for any $v \in (0, \frac{1}{2b}]$, with probability at least $1 - \delta$,

$$\left\| \sum_{\tau=1}^t \Delta_\tau \right\|_2 \leq v \left\| \sum_{\tau=1}^t \mathbb{E}[\Delta_\tau^2 \mid \mathcal{F}_{\tau-1}] \right\|_2 + \frac{1}{v} \log \frac{2d}{\delta}.$$

- ▶ Proved by adapting Bernstein's inequality to matrices via the trace-exponential moment generating function bound $\mathbb{E}_{t-1}[\exp(v\Delta_t)] \preceq \exp(v^2 \mathbb{E}_{t-1}[\Delta_t^2])$, following the technique of Tropp (2012).
- ▶ This is the workhorse concentration tool behind Lemma 3, Lemma 11, and Theorem 5 — it is a standard matrix Bernstein bound, not a contribution specific to this paper.

Source: Sim and Kim (2026), Appendix E.3; adapted from Tropp (2012).

Appendix I: Lemmas 13–14 (pointwise-to-uniform)

Both lemmas convert a pointwise concentration bound (fixed x) into a uniform bound (all $x \in \mathbb{R}$) via a grid of M points with spacing $F(x_k) - F(x_{k-1}) \leq 1/M$, plus a union bound over the $M - 1$ interior grid points.

- **Lemma 14 (scalar).** If $\mathbb{P}(|\hat{F}(x) - F(x)| \leq \epsilon(\delta')) \geq 1 - \delta'$ pointwise, then for $\delta' = \delta/(M - 1)$,

$$\mathbb{P}\left(\sup_{x \in \mathbb{R}} |\hat{F}(x) - F(x)| \leq \epsilon\left(\frac{\delta}{M-1}\right) + \frac{1}{M}\right) \geq 1 - \delta.$$

- **Lemma 13 (vector-weighted).** If $\mathbb{P}(\|(\sum_i \eta_i(x_{i1})v_{i1}, \dots)\|_2 \leq \epsilon(\delta')) \geq 1 - \delta'$ pointwise, then

$$\mathbb{P}\left(\sup_{x \in \mathbb{R}} \left\| \sum_i \eta_i(x)v_i \right\|_2 \leq \epsilon\left(\frac{\delta}{M}\right) + \frac{\sqrt{\sum_i \|v_i\|_2^2}}{M}\right) \geq 1 - \delta.$$

Source: Sim and Kim (2026), Appendix E.4–E.5.

Appendix J: supporting lemmas (15, 16, 19)

- **Lemma 15 (threshold lemma).** For $a > 1/2$ and $b > e^2$,

$$t \geq 4a \left(1 + \log \frac{2a\sqrt{b}}{e} \right) \implies t \geq a \log(bt^2).$$

Used to resolve the implicit T^2 terms in Appendix C.

- **Lemma 16 (Bretagnolle–Huber inequality).** For probability measures $\mathbb{P}_1, \mathbb{P}_2$,

$$1 - \text{TV}(\mathbb{P}_1, \mathbb{P}_2) \geq \frac{1}{2} \exp(-\text{KL}(\mathbb{P}_1, \mathbb{P}_2)).$$

Used in Appendix D to lower-bound the two-point testing error.

- **Lemma 19 (dimension-free martingale bound).** If $\{\eta_s\}$ is σ -sub-Gaussian given $\{\mathcal{F}_s\}$ and X_s is $\mathcal{F}_0$ -measurable, then with probability at least $1 - \delta$,

$$\left\| \sum_{s=1}^t \eta_s X_s \right\|_2 \leq 4\sigma \sqrt{\sum_{s=1}^t \|X_s\|_2^2} \sqrt{2 \log \frac{4t^2}{\delta}}.$$

Used in Appendix E to bound the mixed-estimator error terms.